

Tarrun Pitta

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Education

California State University, Fullerton

Master of Science in Computer Science

Fullerton, CA

08/2024 - 05/2026

Publication: Hardware Trojan Detection with Machine Learning and Power Side-Channels: A Post-Deployment Analysis

Experience

Pacific Life

Newport Beach, CA

Software Engineer

06/2025 - 04/2026

- **Modernized** an enterprise commercial real estate data pipeline containing **stale data**, synchronizing **27,000+ commercial and multifamily property records** representing **\$2B+** in institutional loan assets into **ArcGIS Online**.
- **Engineered** a **full-stack data unification interface** using **React, TypeScript, PostgreSQL, and Redis** to standardize diverse financial data feeds delivered nightly through **Control-M** from multiple institutional sources.
- **Reduced ESRI geocoding API consumption by 95%** and eliminated recurring cloud credit waste by designing a **persistent caching layer** that processed only newly added or modified asset addresses.
- **Resolved** critical asset miscalculations on complex commercial mortgage loans valued up to **\$650M** by implementing **data validation and quarantine workflows** to identify missing fields, malformed balances, and integer overflows.

California State University, Fullerton

Fullerton, CA

Graduate Research Assistant

10/2024 - 06/2025

- **Achieved 98.2% classification accuracy** across **10K+ high-dimensional FPGA side-channel traces** by architecting an end-to-end supervised machine learning pipeline for hardware anomaly detection under variable workloads.
- **Accelerated preprocessing throughput by 4.2x and reduced signal noise by 35%** across **50GB+ telemetry datasets** by vectorizing numerical transformations and feature extraction using **Python, NumPy, and Pandas**.
- **Reduced feature-generation latency by 30%** by building modular preprocessing pipelines that converted raw analog telemetry into normalized, low-latency feature vectors.

Accenture

Hyderabad, India

Software Engineer

08/2022 - 07/2024

- **Developed core REST microservices** with **Java, Spring Boot, and TDD** for Walmart inventory systems, integrating with **React.js** dashboards to ensure real-time store stock visibility.
- **Reduced database query latency by 25%** by profiling **SQL execution plans**, refactoring **12+ Hibernate/JPA mappings**, and optimizing relational indices on **Amazon RDS (AWS)**.
- **Eliminated 95% of malformed transaction records** prior to production ingestion by deploying robust batch pipelines with **Spring Integration, Spring Batch, and relational constraints**.
- **Accelerated deployment velocity by 35%** by provisioning cloud infrastructure with **Terraform**, containerizing services via **Docker**, and deploying to **Kubernetes (K8s)** clusters using **GitHub Actions CI/CD**.
- **Reduced incident triage time by 30%** by configuring **Splunk** logging pipelines and **AWS CloudWatch** alerts to monitor API error spikes and database connection pool health.

Accenture

Hyderabad, India

Software Engineer Co-Op

01/2022 -- 07/2022

- **Implemented 5+ backend API endpoints** using **Java and Spring Boot**, enhancing transaction throughput and data validation against core enterprise services.
- **Diagnosed and resolved 8+ production-critical sprint defects** by debugging REST endpoints and optimizing relational SQL validation scripts.
- **Authored comprehensive technical documentation** and API specs in Swagger/OpenAPI, standardizing onboarding and improving architectural visibility across engineering squads.

Projects

Semantic Code Intelligence Platform | Python, FAISS, BM25, Cross-Encoders, React, FastAPI

GitHub Link

- **Designed a hybrid code-retrieval platform indexing 406K+ lines across 2,464 files** (38K+ semantic chunks) leveraging **FAISS dense retrieval, BM25 lexical search, and Reciprocal Rank Fusion (RRF)**.
- **Achieved 1.24s end-to-end retrieval latency** using incremental indexing, Cross-Encoder reranking, and decoupled **FastAPI** microservices with exact source symbol citations.

Open-SWE | Open-Source Contributor | Python, Docker, LLM Tool Calling, Sandbox Runtimes

- **Expanded autonomous software engineering runtime across 5 merged PRs** by implementing secure **Docker sandbox execution environments** and multi-provider LLM API adapters.

Technical Skills

Languages: Java, Python, TypeScript, JavaScript, SQL, C++, HTML5, CSS3

Backend & Frameworks: Spring Boot, FastAPI, Node.js, Express, Next.js, React.js, Redux, Tailwind CSS

Distributed Systems & Data: REST APIs, Microservices, GraphQL, gRPC, PostgreSQL, MySQL, Redis, Amazon RDS

AI & LLM Engineering: RAG, Vector Search (FAISS), Cross-Encoder Reranking, LangChain, AI Agents, Tool Calling

Cloud & DevOps: AWS (RDS, S3, Lambda, CloudWatch), Kubernetes, Docker, Terraform, CI/CD, GitHub Actions

Observability & Testing: Prometheus, Grafana, OpenTelemetry, Splunk, JUnit, Mockito, Jest, TDD, Agile/Scrum